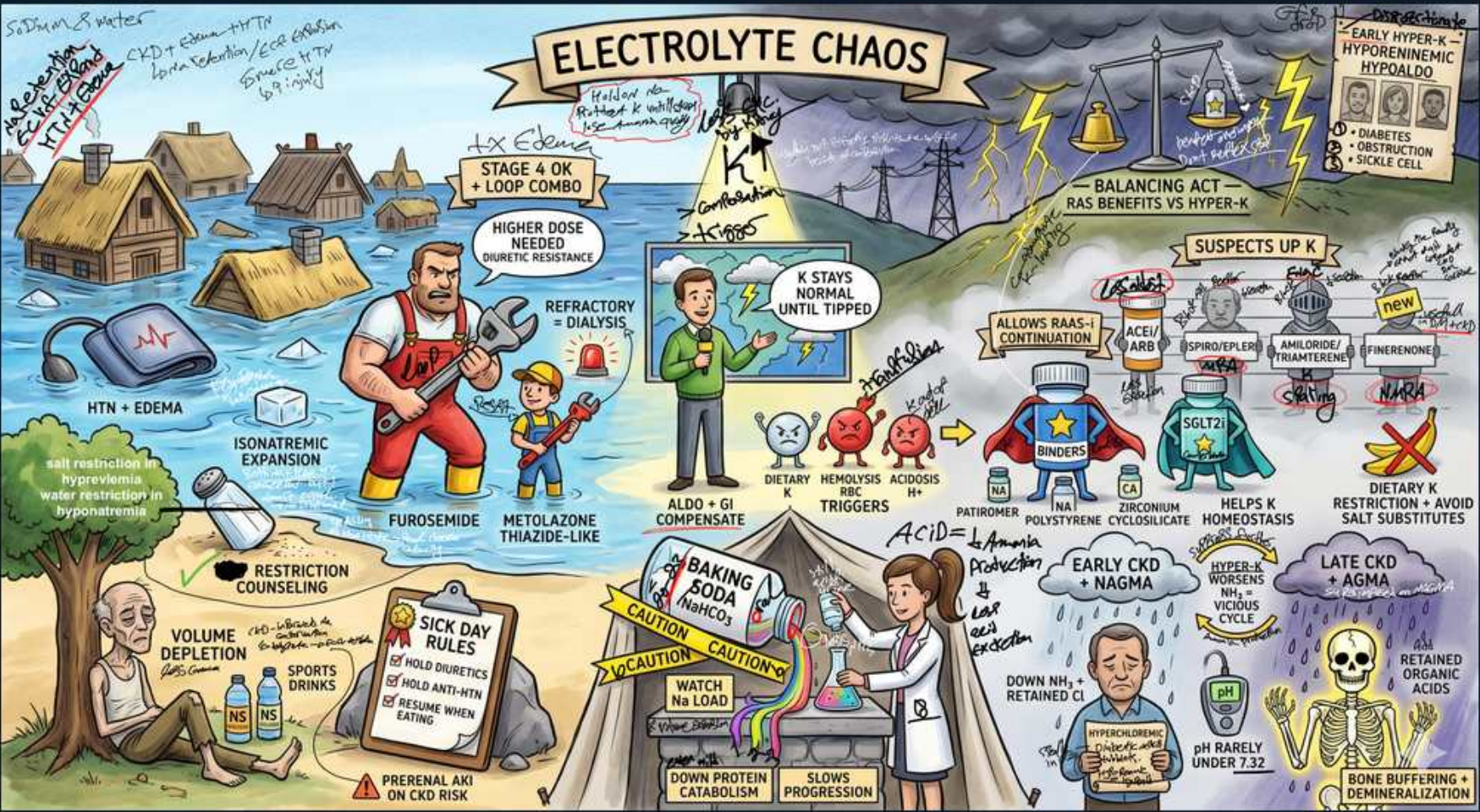


CKD — Electrolyte & Acid-Base Chaos



1. Electrolyte chaos banner

WHAT YOU SEE

Top banner 'Electrolyte chaos'.

WHAT IT ENCODES

As GFR falls below ~30 mL/min (CKD G4–5), the kidney loses its ability to fine-tune water, sodium, potassium, hydrogen ion and ammonium handling, so derangements cluster together. The three boards-classic abnormalities of advanced CKD are hyperkalemia, high-anion-gap metabolic acidosis, and hyperphosphatemia. These tend to appear together late and are the targets of medical management and, ultimately, dialysis.

BOARD TRAP

WRONG to expect symmetric loss of all functions early — CKD preserves K+, acid and phosphate homeostasis until G4–5 via compensatory mechanisms (aldosterone, ammoniogenesis, FGF23). A patient with G2–G3a and a normal K+/HCO3/phosphate is the rule, not a reason to doubt the CKD diagnosis.

2. Flooded village / HTN + edema

WHAT YOU SEE

Flooded huts labeled 'HTN + edema', isotonic expansion.

WHAT IT ENCODES

The failing kidney cannot excrete a sodium load, so Na+ and water are retained in iso-osmotic proportion (isotonic expansion of the extracellular fluid), producing volume overload, hypertension, and peripheral/pulmonary edema. Serum sodium concentration is typically normal because water follows salt. Management is dietary sodium restriction (<2 g/day) plus diuretics; volume overload, not renin, is the dominant driver of CKD hypertension.

BOARD TRAP

WRONG to call this hyponatremia or expect a low serum Na+ — isotonic expansion keeps [Na+] normal even as total-body sodium and water rise. True hyponatremia in CKD instead reflects impaired free-water excretion (excess water relative to salt), a different problem.

3. Furosemide strongman

WHAT YOU SEE

Strongman labeled 'furosemide', 'higher dose needed'.

WHAT IT ENCODES

Loop diuretics (furosemide, bumetanide, torsemide) block the Na-K-2Cl cotransporter in the thick ascending limb and remain the workhorse for volume control in CKD. As GFR falls, less drug is delivered into the tubular lumen and organic-anion competition rises, so higher doses are needed (diuretic resistance) — e.g., IV furosemide 80–200 mg or torsemide, which has better oral bioavailability. Torsemide/bumetanide are favored over furosemide when gut absorption is erratic.

BOARD TRAP

WRONG to abandon loops or switch to a thiazide alone at low GFR. Thiazides lose natriuretic efficacy below eGFR ~30, whereas loops still work and simply need uptitration; the discriminator is that loops act in the thick ascending limb where filtered Na+ load is high.

4. Metolazone / thiazide add-on

WHAT YOU SEE

Caption 'metolazone thiazide-like', 'stage 4 OK + loop combo'.

WHAT IT ENCODES

Metolazone is a thiazide-like agent that blocks the distal convoluted tubule Na-Cl cotransporter; added to a loop it produces sequential nephron blockade, defeating the post-loop distal sodium reabsorption that drives refractory edema. Unlike pure thiazides, metolazone retains efficacy at low GFR, making it the go-to add-on even in CKD G4. Dose ~2.5–10 mg given ~30 min before the loop; watch for profound diuresis, hypokalemia and volume depletion.

BOARD TRAP

WRONG to add metolazone and not monitor — sequential blockade can cause brisk hypokalemia, hyponatremia and prerenal AKI. The combination is potent, so it requires close electrolyte and volume surveillance rather than being a benign add-on.

5. Volume depletion / prerenal AKI

WHAT YOU SEE

Man with sports drinks, 'sick day rules', 'prerenal AKI on CKD'.

WHAT IT ENCODES

Over-diuresis, vomiting/diarrhea, or poor intake during illness causes volume depletion and prerenal acute kidney injury superimposed on CKD (acute-on-chronic), with a rising creatinine, BUN:Cr >20, and bland urine. 'Sick-day rules' instruct patients to temporarily hold diuretics, ACEi/ARB, SGLT2 inhibitors and metformin when acutely ill to protect the kidney. Volume status, not a single creatinine, guides resuscitation.

BOARD TRAP

WRONG to keep diuretics/RAAS blockers running through an acute diarrheal illness — this converts a recoverable prerenal insult into ischemic ATN. The discriminator from intrinsic AKI is the bland sediment and rapid creatinine recovery once volume is restored.

6. K stays normal until tipped

WHAT YOU SEE

Caption 'K stays normal until tipped' near lightning.

WHAT IT ENCODES

Serum potassium is defended until advanced CKD (G4–G5) because remaining nephrons increase distal K+ secretion and the colon ramps up K+ excretion. Hyperkalemia then becomes the lethal danger, with ECG changes progressing from peaked T waves to widened QRS to a sine wave and arrest. Acute hyperK is treated with IV calcium gluconate (membrane stabilization), then insulin+glucose and albuterol (intracellular shift), then loop diuretics/binders/dialysis (removal).

BOARD TRAP

WRONG to attribute a high K+ in early CKD to the renal disease itself — significant hyperkalemia in G1–G3 should prompt a search for a trigger (RAAS blocker, high-K diet, NSAID, acidosis, tissue breakdown, type 4 RTA/hypoaldosteronism), not be dismissed as expected for CKD.

7. Aldo + GI compensate

WHAT YOU SEE

Caption 'aldo + GI compensate'.

WHAT IT ENCODES

Two adaptations keep K⁺ normal until late CKD: aldosterone drives principal-cell K⁺ secretion in surviving nephrons, and the colon upregulates K⁺ excretion (GI compensation can clear up to ~30–50% of a K⁺ load in advanced disease). When these are overwhelmed or blunted (e.g., by ACEi/ARB/MRA, low aldosterone in diabetic type 4 RTA), hyperkalemia emerges.

BOARD TRAP

WRONG to assume the kidney is the only K⁺ excretion route — in advanced CKD the colon becomes a major escape valve, which is why constipation or stopping a potassium binder can unmask hyperkalemia.

8. Hyperkalemia triggers (RAAS, acidosis)

WHAT YOU SEE

Angry K monster, 'dietary, hemolysis, acidosis triggers', 'allows RAAS continuation'.

WHAT IT ENCODES

Common triggers that tip K⁺ up in CKD: high-potassium diet, metabolic acidosis (drives K⁺ out of cells in exchange for H⁺), tissue breakdown/hemolysis, and RAAS inhibitors (ACEi, ARB, MRA, finerenone). Treating the modifiable drivers — correcting acidosis with bicarbonate, restricting dietary K⁺, adding a K⁺ binder — often allows the nephroprotective RAAS blocker to be continued rather than stopped.

BOARD TRAP

WRONG to reflexively discontinue the ACEi/ARB for mild–moderate hyperkalemia. The board-preferred move is to treat the potassium (dietary K⁺ restriction, sodium bicarbonate for acidosis, patiomer/SZC) so the patient keeps the proven renal/cardiac protection; stopping RAAS blockade sacrifices long-term benefit for a manageable lab.

9. K binders (patiomer / SZC)

WHAT YOU SEE

Bottles 'sodium polystyrene', 'zirconium cyclosilicate', 'patiomer'.

WHAT IT ENCODES

Potassium binders exchange K⁺ in the gut for excretion. Newer agents — patiomer (binds K⁺ for Ca²⁺ in the colon) and sodium zirconium cyclosilicate/SZC (a selective inorganic lattice, fast-acting) — are well tolerated for chronic outpatient hyperK and let patients stay on RAAS therapy. Older sodium polystyrene sulfonate (SPS/Kayexalate) is slower and carries a rare risk of intestinal necrosis, especially with sorbitol or postoperatively.

BOARD TRAP

WRONG to choose SPS as a fast emergency treatment for severe hyperkalemia — it is slow and risks colonic necrosis. Acute, ECG-positive hyperK is managed with calcium → insulin/glucose + albuterol → definitive removal (diuretics/dialysis); binders are for chronic control.

10. Avoid salt substitutes

WHAT YOU SEE

Caption 'dietary K restriction + avoid salt substitutes'.

WHAT IT ENCODES

CKD patients are counseled to restrict dietary potassium (high-K culprits: bananas, oranges/orange juice, tomatoes, potatoes, salt substitutes). 'Low-sodium' or 'lite' salt substitutes are largely potassium chloride, so they trade a sodium benefit for a dangerous potassium load. This counseling matters most in G4–5 and in patients on RAAS blockers or with type 4 RTA.

BOARD TRAP

WRONG to recommend a 'salt substitute' to a CKD patient as a heart-healthy swap — these products are potassium chloride and can precipitate life-threatening hyperkalemia. The discriminator is reading the label: KCl, not NaCl.

11. Baking soda / sodium bicarbonate

WHAT YOU SEE

Bag 'baking soda', 'watch Na load', 'slows progression'.

WHAT IT ENCODES

Oral sodium bicarbonate is given to keep serum HCO₃ in the ~22–26 mEq/L range. Correcting metabolic acidosis reduces muscle protein catabolism, improves bone health, lowers potassium, and may slow CKD progression. Typical dosing is sodium bicarbonate 650 mg tablets titrated to the bicarbonate level; veverimer is an investigational non-absorbed alternative.

BOARD TRAP

WRONG to ignore the obligatory sodium load of NaHCO₃ in a volume-overloaded or hypertensive patient — each gram delivers sodium that can worsen edema and BP. Balance acidosis correction against volume; this is the discriminator versus simply maximizing bicarbonate.

12. Early / late metabolic acidosis

WHAT YOU SEE

Clouds 'early = NAGMA', 'late CKD = AGMA', retained acids.

WHAT IT ENCODES

CKD acidosis evolves in two stages. Early (G3) it is a normal-anion-gap (hyperchloremic) metabolic acidosis from impaired ammoniagenesis and reduced acid excretion. Late (G4–5) it becomes a high-anion-gap metabolic acidosis as phosphate, sulfate and other organic acids are retained behind the falling GFR. The anion gap pattern thus tracks CKD severity.

BOARD TRAP

WRONG to expect a high anion gap in early CKD — the G3 pattern is a NAGMA (hyperchloremic). Seeing a wide gap signals advanced disease (G4–5) or a separate cause (lactate, ketones, toxins); using the gap to stage acidosis is the key discriminator.

13. Retained NH₄ / ammonium

WHAT YOU SEE

Caption 'down NH₄ retained', pH note.

WHAT IT ENCODES

Net acid excretion depends largely on urinary ammonium (NH₄⁺); in CKD, fewer functioning nephrons make less ammonia, so the kidney cannot buffer and excrete the daily acid load. Because titratable acid and NH₄⁺ excretion fall, urine pH is often inappropriately near-normal (~5.5) despite systemic acidosis — the defect is quantity of buffer, not the ability to acidify.

BOARD TRAP

WRONG to interpret a non-maximally-acidic urine pH in CKD acidosis as a distal (type 1) RTA. In CKD the urine can still acidify; the problem is reduced ammoniogenesis (low urinary ammonium / low urine anion gap interpretation), not a distal H⁺ secretion defect.

14. Bone buffering / demineralization skeleton

WHAT YOU SEE

Skeleton 'bone buffering — demineralization', 'retained organic acids'.

WHAT IT ENCODES

Chronic acid retention is buffered in part by bone, which releases carbonate and phosphate at the cost of demineralization, worsening CKD—mineral and bone disorder (CKD-MBD) and renal osteodystrophy. This bone buffering is another reason to correct acidosis with bicarbonate. CKD-MBD also involves hyperphosphatemia, low calcitriol, secondary hyperparathyroidism and high FGF23.

BOARD TRAP

WRONG to treat CKD bone disease as ordinary osteoporosis with bisphosphonates first — the underlying drivers are acidosis, phosphate retention, low active vitamin D and secondary hyperPTH. The fix is treating CKD-MBD (phosphate binders, calcitriol/calcimimetics, bicarbonate), not reflexive antiresorptive therapy.
